

Neeloy Chakraborty

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Interests: Embodied AI, Generative AI, Robot Teleoperation, Anomaly and Hallucination Detection, Reinforcement Learning

Skills: Python, PyTorch, PyTorch3D, Ollama, OpenCV, Blender, ROS, ZED Camera, Linux, Git, CARLA, Docker, C/C++

Education:

- **University of Illinois Urbana-Champaign**
PhD in Electrical and Computer Engineering (Exp. May 2026) GPA 3.90/4.00
Master of Science in Electrical and Computer Engineering (2021 – 2023) [Thesis] GPA 3.83/4.00
Bachelor of Science in Computer Engineering (2017 – 2021) [Thesis] GPA 3.75/4.00

Select Experience:

- **Research Assistant in the Human-Centered Autonomy Lab** Fall 2019 – present
Designing intuitive instruction-following embodied AI agents
Developing real-time image generation methods combining depth foundation models and rendering techniques
Detecting failures and hallucinations in LLM generations within multi-agent safety-critical environments
- **PhD Research Intern in Sight Lab at Dolby Laboratories** Summer 2025
Researching greenfield project within the space of creative 3D asset generation
- **Research and Advanced Engineering Intern in Core AI/ML at Ford Motor Company** Summer 2022
Designing sample efficient model-free + model-based RL methods
Outperforming classical PID controllers by 41% in complex autonomous vehicle use case
- **Perception Engineering Intern in Autonomy Team at Brunswick i-Jet Lab** Summer 2021
Localizing swimmers around boats using filtering, tracking, and computer vision techniques
Analyzing functional safety standards practiced at company and presenting findings to global management team
- **Engineering Intern in Global CAD Team at Qualcomm** Summer 2020
Leading design process of generalized data gathering solutions to support Design for Test pipeline
Collaborating and adapting with international teams to consider multiple perspectives

Select Publications, Patents, and Pre-prints:

- **N. Chakraborty**, J. Pohovey, M. Ornik, and K. Driggs-Campbell. “Adaptive Stress Testing Black-Box LLM Planners,” Under review 2025 [Paper]
- **N. Chakraborty***, Y. Fang*, A. Schreiber, T. Ji, Z. Huang, A. Mihigo, C. Wall, A. Almana, and K. Driggs-Campbell. “Towards Real-Time Generation of Delay-Compensated Video Feeds for Outdoor Mobile Robot Teleoperation,” ICRA 2025 [Paper] [Website] [Code] [Dataset] (Leading cross-departmental team of researchers with diverse skillsets)
- K. Balakrishnan, **N. Chakraborty**, and D. Upadhyay. “Model-based Reinforcement Learning,” US20240320505A1 [Patent Pending]
- **N. Chakraborty***, R. Sidhu*, B. Abdullai*, H. Chen*, N. Ravi*, A. Ankur, D. Prasad, and J. Hockenmaier. “BEAST: Building an Embodied Action-prediction System with Trajectory data,” Alexa Prize SimBot Challenge Proceedings 2023 [Paper] [Website] (University team leader and semi-finalist in inaugural Amazon SimBot Challenge)
- **N. Chakraborty**, A. Hasan*, S. Liu*, T. Ji*, W. Liang, D. L. McPherson, and K. Driggs-Campbell. “Structural Attention-based Recurrent Variational Autoencoder for Highway Vehicle Anomaly Detection,” AAMAS 2023 [Paper] [Website] [Code] (Accepted as full paper with 23.3% acceptance rate and received student scholarship)